

Markus Mueller

PhD candidate working on deep generative models (diffusion, flow matching) for tabular and structured data.

Email: mueller@ese.eur.nl
Website: muellermarkus.github.io
GitHub: github.com/muellermarkus
Language: German (native), English (fluent), Dutch (beginner)

Education

ECONOMETRIC INSTITUTE, ERASMUS UNIVERSITY ROTTERDAM

PhD candidate working on deep generative modeling for tabular data, 2022 - expected 2026

Supervisors: Dennis Fok and Kathrin Gruber

BUSINESS DATA SCIENCE / TINBERGEN INSTITUTE

Joint research master programme of the Erasmus University Rotterdam, Vrije Universiteit Amsterdam and the University of Amsterdam

MPhil Business Data Science, with distinction, 2022

UNIVERSITY OF WARWICK

MSc Economics, with distinction, 2018

UNIVERSITY OF STRATHCLYDE

BA (Hons.) Economics, first-class honours (best in cohort), 2017

Research Visits

UNIVERSITY OF AMSTERDAM, AMLAB

Supervisor: Christian A. Naesseth, since Oct 2025

UNIVERSITY OF CAMBRIDGE, VAN DER SCHAAR LAB

Supervisor: Mihaela van der Schaar, Feb - May 2025

Employment

BOOKING.COM

PhD Machine Learning Research Intern, incoming, Summer 2026

Exploiting prior information to improve the target encoding of very high cardinality features using variational inference.

ERASMUS UNIVERSITY ROTTERDAM, ECONOMETRIC INSTITUTE

Doctoral Researcher, Sept 2022 - present

Assisting in various machine learning courses, supervision of bachelor and master theses, administration of coursera course. Research and teaching details are outlined below.

ERASMUS UNIVERSITY ROTTERDAM, BUSINESS ECONOMICS DEPARTMENT

Research Assistant for Bas Donkers, Sept - Dec 2021

Implementing adversarial debiasing methods based on GANs in Julia.

UNIVERSITY OF POTSDAM, CHAIR OF EMPIRICAL ECONOMICS

Full-time Research and Teaching Assistant for Marco Caliendo, 2019/20

Diverse statistical analyses of survey data.

UNIVERSITY OF WARWICK, DEPARTMENT OF ECONOMICS

Research Assistant for Roland Rathelot, 2018

Implementing efficient duration model, partially using C++ and rcpp, to scale model to multi-million data points in administrative database. Data structuring and processing.

Research

Ongoing

Mueller, M. and Naesseth, C. A. (2026) Neural Flow Additive Models. *In preparation.*

We combine Neural Additive Models with flow matching to quantify uncertainty around the feature contributions for probabilistic regression / classification. We use constrained but learnable forward process to adapt to the restrictions imposed by the feature-specific reverse processes. This ensures that the generative process is fully interpretable.

Mueller, M. and Naesseth, C. A. (2026) Improving Diffusion for Panel Data. *In preparation.*

We learn a latent SDE such that the diffusion timestep represents the observational time in a panel data context. This allows us to disentangle latent dynamics over time from time-constant heterogeneity. The efficient reuse of components improves efficiency compared to existing work that disregards the known data structure.

Mueller, M., Kacprzyk, K. and van der Schaar, M. (2025) Prior-Informed Conditional Flow Matching. *In preparation.*

We investigate the possibility of incorporating a set of functional priors based on Bernstein polynomials into the flow matching framework.

Published

Mueller, M., Gruber, K. and Fok, D. (2026) Cascaded Flow Matching for Heterogeneous Tabular Data with Mixed-Type Features. *Accepted at ICML.*

We propose a cascaded model that generates high-resolution information in tabular data based on low-resolution features, including latent variables derived from numerical features. We impose a data-dependent coupling that provably reduces the transport costs. Together with learnable, non-linear guided probability paths, this greatly improves the realism of generated samples.

Mueller, M., Gruber, K. and Fok, D. (2025) Continuous Diffusion for Mixed-Type Tabular Data. ICLR.

Unified framework for continuous and categorical features in diffusion models. We unify both feature types in continuous space and balance their losses to avoid implicit importance weights that impact sample quality.

Open-source codebase adopted by MOSTLY AI for their recent paper.

Mueller, M., Gruber, K. and Fok, D. (2023) Continuous Diffusion for Mixed-Type Tabular Data. NeurIPS Workshop on Synthetic Data Generation with Generative AI.

Very early version of the 2025 paper with some ideas not fletched out.

Theses

Mueller, M. (2022) Leakage in Adversarial Debiasing as a Consequence of a Multi-Directional Bias in Weight Space. *Research Master thesis*, Grade: 9.0/10 (best thesis in cohort), Supervisor: Bas Donkers.

Mueller, M. (2018) Estimating Anticipation and Treatment Effects of Training the Unemployed in France. *Master thesis*, Grade: 82% (one of the top 10 theses in cohort), Supervisor: Roland Rathelot.

Mueller, M. (2017) An Application of Econometric Methods for the Estimation of Causal Effects in Labour Economics. *Bachelor thesis*, Grade: 90% (best thesis in cohort), Supervisor: Eduardo Fé.

Technical Skills

Programming Languages: Python (advanced), R, Julia, C++ (via Rcpp)

ML Frameworks: PyTorch (primary framework for all research projects), JAX (used for simple simulations in research master course)

Research Areas: Deep Generative Models (Diffusion Models, Flow Matching), Probabilistic Machine Learning, Deep Learning for Tabular Data

Computing Infrastructure: High-Performance Computing (Snellius HPC cluster), Docker (vast.ai)

Teaching

ERASMUS UNIVERSITY ROTTERDAM

Machine Learning (postgraduate; conceptualizing and grading assignment), Machine Learning in Econometrics (postgraduate; conceptualizing and grading assignment), Machine Learning in Finance (postgraduate; conceptualizing and grading assignment), Data Science in Context (postgraduate; creating course material and exam, grading), Machine Learning Seminar (undergraduate; conceptualizing content and supervising student groups, grading presentations), Academic Writing (undergraduate; weekly feedback on exercises and grading, giving tutorials)

Bachelor thesis supervision (topics including: transformer-based models and deep learning for tabular data, missing value imputation, uncertainty quantification, denoising autoencoders for tabular data, diffusion models for regression)

Master thesis supervision (topics including: post-hoc auditing of synthetic tabular data, VAE for causal inference)

BUSINESS DATA SCIENCE / TINBERGEN INSTITUTE

Unsupervised Machine Learning (research master; creating and giving coding tutorials in R), Supervised Machine Learning (research master; creating and giving coding tutorials in R), Statistics (research master; giving tutorials for exam preparation)

UNIVERSITY OF POTSDAM

Machine Learning and Economics (postgraduate; course setup and creation of teaching material, leading seminar discussions), Statistics (undergraduate, online; creation of tutorial videos), Advanced Microeconometrics (postgraduate; lecturer), Applied Public Policy Evaluation (postgraduate; lecturer)

Professional Services

Reviewer: AAAI Good-Data Workshop, 2025; ICLR, 2025; NeurIPS 2026 (upcoming)

Joint Organizer: Lecture Series on Causal Inference and Machine Learning by Guido W. Imbens, University of Potsdam, 2019; 6th PhD Workshop on Empirical Economics, University of Potsdam, 2019; Econometric PhD Internal Conference, Erasmus University Rotterdam, 2025

Talks/Presentations

Invited talk: Cambridge Center for AI in Medicine Seminar, 2024

Erasmus Center for Data Analytic Insights Social AI Seminar, 2023

Research Group Seminar (Statistics and Computer Science) at the Econometric Institute of the Erasmus University Rotterdam, 2022, 2023

PhD Lunch Seminar, Erasmus School of Economics, Erasmus University Rotterdam, 2022

Selected Honors

2nd place (average performance) in the MOSTLY AI synthetic data generation challenge, 2025

Nominated for the ODISSEI Open Science Award 2025 and the Open Research Award at the Erasmus University Rotterdam 2024, 2025

Full Scholarship (merit-based) for MPhil Business Data Science, 2021

Partial Scholarship for MPhil Business Data Science, 2020

Top 10 Dissertation Award, Department of Economics, University of Warwick, 2018

Sir Charles Carter Prize for outstanding performance (highest average grade in cohort), University of Strathclyde, 2017

Department of Economics Prize for the best dissertation in Economics, University of Strathclyde, 2017

Grants

NWO/SURF EINF grant (~ 2 000 Euros) for computing resources (reference code: EINF-13805)

NWO/SURF EINF grant (~ 12 000 Euros) for computing resources (reference code: EINF-7437)

ERIM funding of travel costs (1 500 Euros)

ERIM funding of research visit (3 000 Euros)

Summer Schools and Workshops

High-Performance Computing Tutorial by SURF, 2025

Cambridge ELLIS Unit Summer School on Probabilistic Machine Learning, 2023

Workshop on *Bayesian Conjoint Analysis* by Thomas Otter, 2021

Summer school on *Artificial Intelligence without Bias* by the NoBIAS Research Group, 2021

Workshop on *Machine Learning - An Applied Econometric Approach* by Jann Spiess, 2019

Last updated: April, 2026